MLFF Architecture Revision 46

PERF-P1 shared exact selection and linear-memory DATA7 closure

mdstats project

2026-08-15

1 Scope

Revision 46 closes bounded `PERF-P1`. TARGET-DATA2C and DATA7 now share one exact deterministic FPS state, TARGET-DATA2C builds its fused matrix directly into one preallocated FP64 array, nested TARGET-DATA2B coverage can advance one family state across rungs, and DATA7 selected-neighbor coverage no longer keeps a dense KxK persistent matrix.

The release is `mdstats 0.20.180a0`. `PERF-P2` is next.

2 Scientific boundary

PERF-P1 is Authority Class E. Existing scientific schemas and digests do not change. The release preserves:

- quota-selected prefixes;
- exact FPS order and lexical tie resolution;
- fused-selector bytes;
- TARGET-DATA2B family/rung statistics;
- Stage-A-relevant coverage reports; and
- DATA7 selected membership and nearest-other-selected coverage.

3 Exact FPS state

`ExactFPSState` owns immutable selector geometry plus reusable execution state: row norms, lexical UID rank, selected mask/rank, minimum squared distance, and selected order. Appending x_j updates candidates from

$$\|x_i - x_j\|_2^2 = \|x_i\|_2^2 + \|x_j\|_2^2 - 2x_i \cdot x_j.$$

Quota selection initializes this state and exact FPS continues it directly. Gonzalez's farthest-first traversal is the standard algorithmic antecedent [1], but mdstats authority remains its frozen deterministic project order and tie policy.

4 Shared coverage state

`score_target_nested_subsets_coverage` processes one family across nested rungs, updates nearest-selected reference distance only against newly added selected points, and releases the family before moving to the next. Exact cKDTree queries use the campaign CPU-budget worker count rather than a hard-coded serial setting [2]. Existing one-dimensional Wasserstein statistics remain unchanged [3].

5 DATA7 memory correction

The former incremental DATA7 path still retained a dense KxK selected-distance matrix. Revision 46 replaces it with one FP64 minimum-squared-distance vector of length K. Bounded old-new and new-new pair blocks update both endpoints, so all exact pair distances needed for nearest-other-selected coverage remain considered.

At K=8192, persistent state changes from

$$8K^2 = 512 \text{ MiB}$$

to

$$8K = 64 \text{ KiB}.$$

6 Qualification

The supplied-data oracle is the PERF-P0 native reference: 37,633 frames, eight coverage families, and a 37,633x50 fused selector matrix. Full selector bytes, FPS order through K=1024, and coverage-report digests at 128/256/512/1024 are exactly unchanged.

Benchmark	Legacy	PERF-P1	Result
Full-reference FPS K=1024	1.550 s	0.657 s	**57.61% faster**
Wide FPS 4000x128, K=512	0.182 s	0.049 s	**73.25% faster**
DATA7 K=8192 wall	1.954 s	0.506 s	**74.09% faster**
DATA7 K=8192 peak RSS	1149.89 MiB	637.22 MiB	**44.58% lower**
DATA7 persistent state	512 MiB	64 KiB	**8192x smaller**

Four-rung progressive coverage is exact but its median wall time is 10.29% slower on this host. This is recorded as a PERF-P2+ optimization opportunity, not represented as a speedup.

7 Evidence and next gate

Normative/evidence surfaces are:

- `docs/specs/training_data/mlff_perf_p1_shared_exact_selection_spec.{md,pdf}`;
- `audits/analysis/mlff_perf_p1_lta_cloud_cpu_2026-08-15.{json,md,pdf}`;
- `release/MLFF_PERF_P1_QUALIFICATION_0.20.180a0.json`; and
- canonical architecture revision 46.

`PERF-P2` may change TARGET-DATA2C ladder authority to progressive/lazy materialization only under its separate Authority Class A contract. PERF-P1's exact workspace and frozen scientific oracle are its execution baseline.

8 References

[1] T. F. Gonzalez, "Clustering to Minimize the Maximum Intercluster Distance," *Theoretical Computer Science* **38**, 293–306 (1985). DOI: 10.1016/0304-3975(85)90224-5.

[2] SciPy developers, "scipy.spatial.cKDTree.query," SciPy reference documentation. Available at: https://docs.scipy.org/doc/scipy/reference/generated/scipy.spatial.cKDTree.query.html (accessed 2026-08-15).

[3] SciPy developers, "scipy.stats.wasserstein_distance," SciPy reference documentation. Available at: https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.wasserstein_distance.html (accessed 2026-08-15).